

Comparative Analysis of Seeding Strategies in K-means Clustering

Stability and Clustering Quality across
Low- and High-Dimensional Data

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April 28, 2026

Abstract

K-means is a widely used unsupervised learning algorithm for grouping similar observations into clusters. Although the algorithm is simple and efficient, its final result depends strongly on the initial positions of the cluster centroids. A poor initialization can lead to a weak local solution, while a better initialization can improve the quality and stability of the final clustering.

This report studies the effect of centroid initialization by comparing Random Seeding and K-means++. The experiments are performed on two different datasets: a Synthetic Blobs dataset and the MNIST handwritten digit dataset. The first dataset provides a low-dimensional and controlled environment with overlapping clusters, while MNIST provides a more complex high-dimensional setting with 784 pixel features.

Before clustering, the data are standardized so that all features have a similar influence on distance calculations. The two initialization methods are evaluated using inertia, mean inertia, and standard deviation across multiple runs. These measures are used to study both clustering quality and variability.

The results show that the two methods perform similarly on the Synthetic Blobs dataset. This is mainly because the clusters overlap and random initialization can still produce reasonable starting centroids. On MNIST, K-means++ gives a slightly lower mean inertia and a slightly lower standard deviation. This suggests that careful centroid selection becomes more useful when the data are high-dimensional and more difficult to cluster.

Overall, the study shows that initialization is an important part of K-means. The results also show that the best initialization strategy depends on the structure and complexity of the dataset.

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1 Introduction

Clustering is an important technique in unsupervised machine learning. Unlike supervised learning, clustering does not require labeled examples. Instead, the goal is to discover groups of observations that are similar to each other. This makes clustering useful for finding hidden patterns and structures in data.

K-means is one of the most common clustering algorithms because it is simple, efficient, and relatively easy to implement. The algorithm divides the observations into a predefined number of clusters. Each observation is assigned to the closest centroid, and the centroids are then updated according to the observations assigned to them.

However, K-means has an important limitation. The final solution can depend on the initial position of the centroids. Different initial centroids can lead the algorithm to different local solutions. Therefore, two runs of K-means on exactly the same dataset can produce different results.

This issue is especially important when the data contain many features or when the clusters have a complicated structure. If the starting centroids are poorly placed, some groups may not be represented well. This can lead to higher inertia and less stable results.

This report focuses on this initialization problem. Two strategies are compared:

- **Random Seeding**, where the initial centroids are selected randomly from the observations;
- **K-means++**, where the first centroid is selected randomly and the remaining centroids are selected using a distance-based probability.

The main purpose of the study is to understand whether the more careful initialization of K-means++ provides a practical improvement over random initialization.

Two datasets are used. The first is a Synthetic Blobs dataset with overlapping clusters. The second is MNIST, where each handwritten digit is represented using 784 pixel features. This allows the comparison to cover both a relatively simple low-dimensional setting and a more challenging high-dimensional setting.

The main research questions are:

1. How does centroid initialization affect K-means clustering?
2. Are Random Seeding and K-means++ equally effective on different types of data?
3. Does K-means++ provide more stable results on high-dimensional data?

The report first explains the K-means objective and the two initialization methods. It then describes the datasets and preprocessing procedure. Finally, the experimental results are presented and discussed in detail.

2 Theoretical Framework

2.1 The K-means Optimization Problem

The objective of K-means is to divide n observations into k clusters such that observations inside the same cluster are close to their cluster centroid.

Let the clusters be

$$S = \{S_1, S_2, \dots, S_k\},$$

and let μ_i be the centroid of cluster S_i . K-means minimizes the within-cluster sum of squared distances, also called inertia:

$$J = \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2. \quad (1)$$

Here, x represents a data point and μ_i represents the centroid of the cluster assigned to that point. A smaller value of J means that the observations are, on average, closer to their assigned centroids. Therefore, lower inertia generally indicates more compact clusters.

The standard K-means procedure contains two main repeated steps:

1. Assign each data point to its nearest centroid.
2. Recalculate each centroid using the mean of the points assigned to that cluster.

These steps are repeated until the centroids change very little or a maximum number of iterations is reached.

The algorithm is efficient, but it does not guarantee the global optimum of the objective function. Instead, it usually converges to a local minimum. This is why initialization is important.

2.2 Random Seeding

Random Seeding is the simplest initialization strategy. The algorithm randomly selects k observations from the dataset and uses them as the initial centroids.

The main advantage of this approach is its simplicity. It is fast and easy to implement. However, the selected points may be close to one another. In this case, some parts of the data may not have a suitable initial centroid.

For example, if two initial centroids are selected from the same natural group, another group may have no centroid near it. K-means can sometimes correct this during later iterations, but the final result can still be worse than the result obtained from a better initialization.

Random Seeding is also sensitive to randomness. Repeating the algorithm with different random starting points can therefore produce different inertia values.

2.3 K-means++ Initialization

K-means++ improves the initial centroid selection by trying to place centroids farther apart.

The first centroid is selected randomly. For every later centroid, the distance from each data point to its nearest already selected centroid is calculated. Points that are farther from the existing centroids receive a higher probability of being selected.

The basic idea can be expressed as

$$P(x) = \frac{D(x)^2}{\sum_{z \in X} D(z)^2}, \quad (2)$$

where $D(x)$ is the distance from point x to its nearest selected centroid.

This strategy reduces the probability that all initial centroids are placed in the same region of the data space. It therefore gives K-means a more balanced starting point.

The main difference between the two methods can be summarized as follows.

Property	Random Seeding	K-means++
Centroid selection	Random selection	Distance-based selection
Main advantage	Simple and fast	Better initial coverage of data
Main weakness	Can select poor starting points	Slightly more complex initialization
Expected stability	More sensitive to random starting points	Usually more stable
Usefulness	Suitable for simple problems	Especially useful for complex data

Table 1: Comparison of the two centroid initialization strategies.

3 Methodology

3.1 Experimental Design

The experiment compares Random Seeding and K-means++ under the same general K-means procedure. The comparison is performed on two datasets with different characteristics.

The first dataset is Synthetic Blobs. It contains 1,000 observations distributed around eight cluster centers. A cluster standard deviation of 2.5 is used, which creates overlapping groups. This makes the dataset useful for studying whether initialization has a strong effect when the cluster boundaries are not clearly separated.

The second dataset is MNIST. In this experiment, 2,000 observations are used. Each handwritten digit is represented by 784 pixel features. This creates a much higher-dimensional problem than the Synthetic Blobs dataset.

The main characteristics of the datasets are shown in Table 2.

Dataset	Size and Dimension	Purpose
Synthetic Blobs	1,000 samples, 8 cluster centers	Controlled low-dimensional experiment with overlapping clusters
MNIST	2,000 samples, 784 features	High-dimensional experiment using handwritten digit images

Table 2: Datasets used in the experimental study.

3.2 Data Preprocessing

Preprocessing is important because K-means is based on distance calculations. If one feature has much larger values than another feature, it can have a much stronger effect on the distance.

To avoid this problem, StandardScaler is used before clustering. The standardization formula is

$$z = \frac{x - \mu}{\sigma}, \quad (3)$$

where x is the original feature value, μ is the mean of the feature, and σ is its standard deviation.

After standardization, each feature has approximately zero mean and unit standard deviation.

This step is particularly important for MNIST. Pixel values originally range from 0 to 255, so scaling makes the distance calculation more balanced across all 784 dimensions.

3.3 Evaluation Metric

The main performance measure is inertia. It is calculated using Equation 1. Lower inertia means that the data points are closer to their assigned centroids.

However, a single K-means run is not enough to evaluate initialization. The algorithm contains randomness, especially when Random Seeding is used. Therefore, the experiments are repeated multiple times.

For each initialization method, two summary statistics are considered:

- **Mean inertia:** shows the average clustering quality across the experimental runs.
- **Standard deviation:** shows how much the inertia changes from one run to another.

A lower mean inertia is generally preferred because it represents more compact clusters. A lower standard deviation is also useful because it indicates that the method produces more consistent results.

Therefore, the analysis does not only ask which method gives a lower inertia. It also asks which method gives more stable results across different runs.

3.4 Experimental Procedure

The experiment follows the same general procedure for both datasets:

1. Load the dataset.
2. Standardize the features.
3. Select the required number of clusters.
4. Initialize centroids using Random Seeding.
5. Run K-means until convergence.
6. Calculate the final inertia.
7. Repeat the process for multiple trials.
8. Repeat the same experiment using K-means++.
9. Compare the mean inertia and standard deviation.

Using the same general procedure for both methods makes the comparison fair. The main difference between the experiments is therefore the initialization strategy.

4 Results and Analysis

4.1 Synthetic Blobs Dataset

The first experiment uses the Synthetic Blobs dataset. The clusters overlap because the standard deviation is set to 2.5. This creates a useful test case because even a random initialization can sometimes place the centroids in reasonable positions.

Figure 1 presents the distribution and variability of the obtained inertia values.

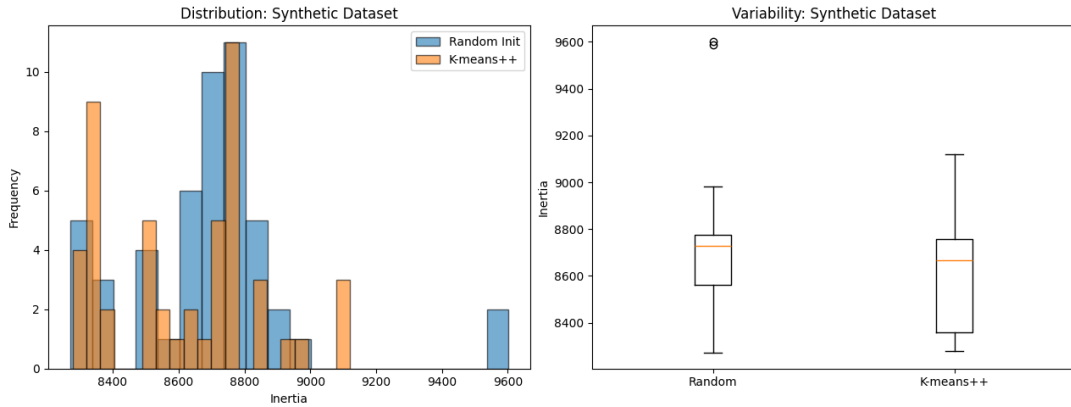


Figure 1: Inertia distribution and variability for the Synthetic Blobs dataset.

The numerical results are shown in Table 3.

Initialization Method	Mean Inertia	Standard Deviation
Random Seeding	8631.71	228.84
K-means++	8679.65	279.37

Table 3: Results over 50 trials on the Synthetic Blobs dataset.

The mean inertia values are very close. Random Seeding obtains a mean inertia of 8631.71, while K-means++ obtains 8679.65. The difference is small compared with the overall scale of the inertia values.

Interestingly, Random Seeding has a slightly lower mean inertia in this experiment. This does not mean that Random Seeding is generally better than K-means++. The result is specific to this dataset and these experimental runs.

The standard deviation also gives an important result. Random Seeding has a standard deviation of 228.84, while K-means++ has a standard deviation of 279.37. Therefore, Random Seeding shows slightly less variation in these particular Synthetic Blobs experiments.

The main explanation is the structure of the dataset. The clusters overlap, but the problem is still relatively controlled and low-dimensional. Even when the initial centroids are selected randomly, there is a reasonable chance that they are placed close to useful regions.

This reduces the advantage of K-means++. In other words, the more careful initialization strategy is not always necessary when the data structure is simple enough for random starting points to work well.

The result is important because it shows that K-means++ should not be described as automatically better in every situation. Its main value is to reduce the risk of poor initialization, especially when the data are more complex.

4.2 MNIST Dataset

The second experiment uses the MNIST dataset. Unlike the Synthetic Blobs dataset, MNIST is high-dimensional. Each image is represented by 784 pixel features. This makes the clustering problem more difficult.

Figure 2 shows the distribution and variability of inertia for the two initialization methods.

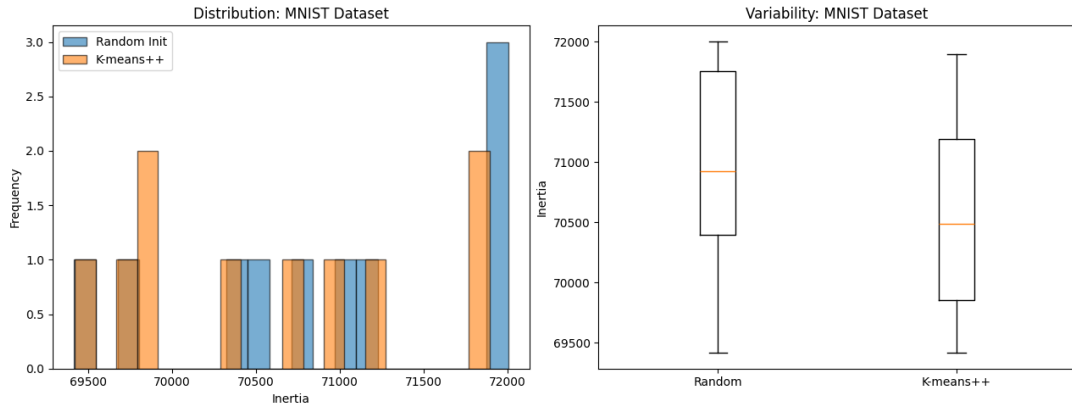


Figure 2: Inertia distribution and variability for the MNIST dataset.

The numerical results are given in Table 4.

Initialization Method	Mean Inertia	Standard Deviation
Random Seeding	70900.91	872.88
K-means++	70573.76	846.36

Table 4: Comparative results for the MNIST dataset.

The MNIST results show a clearer advantage for K-means++. The mean inertia decreases from 70900.91 with Random Seeding to 70573.76 with K-means++. This means that K-means++ produces slightly more compact clusters in this experiment.

The standard deviation also decreases from 872.88 to 846.36. This is a small but useful improvement in consistency. It indicates that the results obtained using K-means++ are slightly less variable across the experimental runs.

The difference is understandable because MNIST contains 784 features for every observation. In such a high-dimensional space, the choice of initial centroids can have a stronger influence on the path followed by K-means.

Random Seeding may select starting points that do not represent the different regions of the data well. K-means++ tries to spread the initial centroids across the data space. This can provide a better starting position for the later K-means iterations.

The results therefore support the main expectation of the study: K-means++ can be more useful when the dataset is high-dimensional and more complex.

4.3 Overall Comparison

The results from both datasets are summarized in Table 5.

Dataset	Better Mean Inertia	Lower Variability	Main Observation
Synthetic Blobs	Random Seeding	Random Seeding	Both methods are very similar
MNIST	K-means++	K-means++	K-means++ gives a small improvement

Table 5: Overall comparison of the two initialization methods.

The two experiments show that initialization performance depends on the dataset.

For Synthetic Blobs, Random Seeding performs slightly better according to both reported statistics. However, the difference is small. This suggests that the structure of the dataset makes the choice of initialization less important.

For MNIST, K-means++ performs better according to both mean inertia and standard deviation. The improvement is again not very large, but the direction of the result is consistent with the idea that careful initialization becomes more valuable for difficult high-dimensional data.

The most important conclusion from this comparison is therefore not that one method always wins. Instead, the results show that the effect of initialization is connected to the structure and complexity of the dataset.

5 Discussion

The experiments provide a clear comparison between Random Seeding and K-means++. The results also show why it is important to evaluate an algorithm on more than one dataset.

On the Synthetic Blobs dataset, the difference between the two methods is very small. The mean inertia values are close, and Random Seeding even has a slightly lower value. The same method also has a lower standard deviation in these experiments.

This result can be explained by the overlapping structure of the Synthetic Blobs dataset. Although the clusters are not perfectly separated, the data are still low-dimensional and controlled. Random centroids can therefore be placed in useful positions by chance.

The MNIST experiment gives a different picture. K-means++ has a lower mean inertia and a lower standard deviation. The difference is not large, but both measures move in the same direction. This suggests that K-means++ can produce slightly more compact and more consistent clusters in a high-dimensional setting.

There is also an important methodological point. Inertia alone does not fully describe clustering quality. A lower inertia means that points are closer to their assigned centroids, but it does not necessarily mean that the clusters correspond perfectly to the actual digit classes in MNIST. The experiment therefore focuses on the effect of initialization on the K-means objective rather than on classification accuracy.

Another important point is the role of preprocessing. K-means is based on distances, so the scale of the features directly affects the clustering process. Standardization makes the contribution of the features more balanced. This is particularly relevant for MNIST because each observation has many pixel features.

The results also demonstrate the importance of repeated experiments. Because the initial centroids can change from one run to another, a single run can give a misleading impression of performance. Looking at the mean and standard deviation provides a better picture of the typical result and the amount of variability.

Overall, K-means++ appears to be a useful default initialization method, especially when the data are high-dimensional or have a more complex structure. However, the Synthetic Blobs experiment shows that its advantage is not guaranteed. For simpler datasets, Random Seeding may perform similarly and can even give a slightly better result in a particular set of runs.

Therefore, initialization should be considered as one part of the overall clustering procedure. Dataset structure, preprocessing, number of clusters, and the number of repeated runs can all influence the final result.

6 Limitations

Although the experiment provides useful evidence, several limitations should be considered.

First, only two datasets are used. Synthetic Blobs provides a controlled environment, while MNIST provides a high-dimensional environment. However, these two datasets cannot represent every possible type of clustering problem.

Second, the Synthetic Blobs dataset contains overlapping clusters. This is useful for testing the methods, but it also makes the difference between the initialization strategies less visible.

Third, the analysis is mainly based on inertia and its standard deviation. These measures are useful for studying the K-means objective and stability, but they do not directly measure whether the discovered clusters correspond to meaningful semantic groups.

Fourth, the results depend on the selected experimental settings, including the number of clusters and the number of trials. Different settings may produce different numerical results.

Finally, the study focuses only on Random Seeding and K-means++. Other initialization strategies and other clustering algorithms could provide additional comparisons.

These limitations do not invalidate the results. Instead, they show that the conclusions should be interpreted within the experimental setting used in this report.

7 Conclusion

This report examined how centroid initialization affects the performance and stability of K-means clustering. Random Seeding and K-means++ were compared using a Synthetic Blobs dataset and the MNIST handwritten digit dataset.

The Synthetic Blobs experiment showed very similar performance for the two methods. Random Seeding achieved a mean inertia of 8631.71 and a standard deviation of 228.84, while K-means++ achieved a mean inertia of 8679.65 and a standard deviation of 279.37. The small difference can be explained by the overlapping but relatively simple structure of the dataset.

The MNIST experiment produced a different result. Random Seeding obtained a mean inertia of 70900.91 with a standard deviation of 872.88. K-means++ obtained a lower mean inertia of 70573.76 and a lower standard deviation of 846.36. This suggests that K-means++ produced slightly more compact and consistent clusters in the high-dimensional setting.

The results therefore show that the usefulness of initialization depends on the dataset. Random Seeding can be sufficient for relatively simple data, while K-means++ can provide a more reliable starting point for complex and high-dimensional data.

The study also confirms the importance of preprocessing. Standardizing the features helps prevent variables with larger scales from dominating the distance calculations. This is especially important when working with high-dimensional image data such as MNIST.

A final important observation is that K-means++ should not be considered universally superior based on a single experiment. In this study, its advantage is small on Synthetic Blobs but more visible on MNIST. The main benefit of K-means++ is its ability to reduce the risk of poor initial centroid placement.

Overall, the experiment provides a practical understanding of why initialization matters in K-means. It also shows that a good clustering method should be evaluated across different types of data rather than on only one dataset.

A Experimental Summary

This appendix provides a short summary of the experimental settings used in the report. It is included to make the experimental procedure easy to review without presenting the implementation code.

Setting	Description
Clustering algorithm	K-means implemented for the experiment
Initialization methods	Random Seeding and K-means++
Synthetic dataset	1,000 samples, 8 cluster centers
Synthetic cluster spread	Standard deviation of 2.5
MNIST subset	2,000 observations
MNIST dimensionality	784 pixel features per observation
Preprocessing	StandardScaler standardization
Synthetic evaluation	50 experimental trials
Main metric	Inertia
Stability measure	Standard deviation of inertia

Table 6: Summary of the main experimental settings.

The experimental comparison was designed so that both initialization methods were evaluated using the same general K-means procedure. The main difference was the way the initial centroids were selected.

The Synthetic Blobs dataset was used to study a controlled, low-dimensional setting. The MNIST dataset was used to study a high-dimensional setting. This design makes it possible to observe whether the effect of initialization changes when the complexity of the data increases.

The final comparison was based on the average inertia and its variability. This provides a simple and clear way to study both clustering compactness and consistency across repeated runs.

References

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Date: April 28, 2026